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# LECTURE 19: Regenerative Rankine Cycle
# Script: rankine-regen.ipynb
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# Description:
# This notebook performs a thermodynamic analysis of a
↳ regenerative Rankine
# cycle with one open feedwater heater (OFWH). It calculates the
↳ states, work,
# heat, and thermal efficiency of the cycle based on user inputs
↳ for boiler
# pressure, boiler temperature, extraction pressure, condenser
↳ pressure,
# and isentropic efficiency. The notebook uses CoolProp to
↳ obtain fluid
# properties and ipywidgets to create an interactive interface
↳ for exploring
# how different operating conditions affect cycle performance.
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# Install necessary packages if you don't have them
# !pip install CoolProp ipywidgets

import CoolProp.CoolProp as CP
import ipywidgets as widgets
from IPython.display import display, clear_output

def solve_regenerative_rankine(P_boiler_bar, T_boiler_C,
↳ P_extract_bar, P_cond_bar, eta):
    """
    Calculates the states and efficiency of a Regenerative
↳ Rankine Cycle
    with one Open Feedwater Heater (OFWH). Uses IUPAC sign
↳ convention.
    """
    fluid = 'Water'

    # Check for non-physical pressure selections from sliders
    if P_extract_bar >= P_boiler_bar or P_extract_bar <=
↳ P_cond_bar:
        print("Error: Extraction pressure must be between Boiler
↳ and Condenser pressures.")
        return

    # Convert pressures to Pa and Temp to K for CoolProp

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P1 = P_boiler_bar * 1e5
T1 = T_boiler_C + 273.15
P_ex = P_extract_bar * 1e5
P_cond = P_cond_bar * 1e5

try:
    # --- STEP 1: Boiler Out / HP Turbine In (State 5) ---
    # Properties come from CoolProp calls
    H1 = CP.PropsSI('H', 'P', P1, 'T', T1, fluid) / 1000 #
↪ kJ/kg
    S1 = CP.PropsSI('S', 'P', P1, 'T', T1, fluid) / 1000 #
↪ kJ/kg-K

    # --- STEP 2: HP Turbine Out / Extraction (State 2) ---
    # Ideal
    S2_ideal = S1
    H2_ideal = CP.PropsSI('H', 'P', P_ex, 'S', S2_ideal *
↪ 1000, fluid) / 1000
    W_HP_rev = H2_ideal - H1 # IUPAC: Negative
    # Actual
    W_HP = W_HP_rev * eta
    H2 = H1 + W_HP
    S2_actual = CP.PropsSI('S', 'P', P_ex, 'H', H2 * 1000,
↪ fluid) / 1000

    # --- STEP 3: LP Turbine Out / Condenser In (State 3)
    ↪ ---
    # Ideal
    S3_ideal = S2_actual
    H3_ideal = CP.PropsSI('H', 'P', P_cond, 'S', S3_ideal *
↪ 1000, fluid) / 1000
    W_LP_rev = H3_ideal - H2 # IUPAC: Negative
    # Actual
    W_LP = W_LP_rev * eta
    H3 = H2 + W_LP

    # --- STEP 4: Condenser Out / Pump 1 In (State 4) ---
    H4 = CP.PropsSI('H', 'P', P_cond, 'Q', 0, fluid) / 1000
↪ # Sat Liquid
    D4 = CP.PropsSI('D', 'P', P_cond, 'Q', 0, fluid) #
↪ kg/m^3
    V4 = 1 / D4 # m^3/kg

    # --- STEP 5: Pump 1 Out / OFWH In (State 5) ---
    W_P1_rev = V4 * (P_ex - P_cond) / 1000 # kJ/kg. IUPAC:
↪ Positive
    W_P1 = W_P1_rev / eta
    H5 = H4 + W_P1

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# --- STEP 6: OFWH Out / Pump 2 In (State 6) ---
H6 = CP.PropsSI('H', 'P', P_ex, 'Q', 0, fluid) / 1000 #
↪ Sat Liquid
D6 = CP.PropsSI('D', 'P', P_ex, 'Q', 0, fluid)
V6 = 1 / D6

# --- MASS BALANCE: Open Feedwater Heater ---
# y*H2 + (1-y)*H5 = H6
y = (H6 - H5) / (H2 - H5)

# --- STEP 7: Pump 2 Out / Boiler In (State 7) ---
W_P2_rev = V6 * (P1 - P_ex) / 1000 # kJ/kg
W_P2 = W_P2_rev / eta
H7 = H6 + W_P2

# --- CYCLE PERFORMANCE ---
W_turb_total = W_HP + (1 - y) * W_LP
W_pump_total = (1 - y) * W_P1 + W_P2
W_net = W_turb_total + W_pump_total

Q_H = H1 - H7

thermal_efficiency = -W_net / Q_H

# --- PRINT RESULTS ---
print("-" * 50)
print(f"REGENERATIVE RANKINE CYCLE ANALYSIS")
print("-" * 50)
print(f"Extraction Fraction (y):      {y:.4f}")
↪   ({y*100:.1f}%)")
print("-" * 50)
print(f"HP Turbine Work (per kg):      {W_HP:.2f} kJ/kg")
print(f"LP Turbine Work (per kg):      {W_LP:.2f} kJ/kg")
print(f"Total Turbine Work:              {W_turb_total:.2f}")
↪   kJ/kg")
print(f"Total Pump Work:                  {W_pump_total:.2f}")
↪   kJ/kg")
print(f"Net Work (W_net):                 {W_net:.2f} kJ/kg")
print(f"Heat Added (Q_H):                {Q_H:.2f} kJ/kg")
print("-" * 50)
print(f"THERMAL EFFICIENCY:               {thermal_efficiency*100:.2f} %")
↪   ({e}))")
print("-" * 50)

except Exception as e:
    print(f"Fluid property error. Trying to calculate")
    ↪   outside valid region. ({e}))")

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# --- CREATE INTERACTIVE WIDGETS TO INPUT OPERATING PARAMETERS
↪ ---
style = {'description_width': 'initial'}

slider_P_boiler = widgets.FloatSlider(value=60.0, min=20.0,
↪ max=150.0, step=1.0, description='Boiler Pressure (bar):',
↪ style=style)
slider_T_boiler = widgets.FloatSlider(value=500.0, min=300.0,
↪ max=600.0, step=5.0, description='Boiler Temp (°C):',
↪ style=style)
slider_P_extract = widgets.FloatSlider(value=10.0, min=2.0,
↪ max=30.0, step=1.0, description='Extraction Pressure
↪ (bar):', style=style)
slider_P_cond = widgets.FloatSlider(value=1.0, min=0.05,
↪ max=2.0, step=0.05, description='Condenser Pressure (bar):',
↪ style=style)
slider_eta = widgets.FloatSlider(value=0.75, min=0.5, max=1.0,
↪ step=0.01, description='Isentropic Efficiency (η):',
↪ style=style)

# Group them into a nice vertical layout
ui = widgets.VBox([slider_P_boiler, slider_T_boiler,
↪ slider_P_extract, slider_P_cond, slider_eta])

# Define the output display connection
out = widgets.interactive_output(solve_regenerative_rankine, {
    'P_boiler_bar': slider_P_boiler,
    'T_boiler_C': slider_T_boiler,
    'P_extract_bar': slider_P_extract,
    'P_cond_bar': slider_P_cond,
    'eta': slider_eta
})

# Display the interface
display(ui, out)

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VBox(children=(FloatSlider(value=60.0, description='Boiler Pressure (bar):'

Output()

**Figure 1.** Modified Rankine Cycle with open feedwater heater.